Generative Modelling

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November 13, 2026

Outline

Graphical models
- Random variables and probabilities
- Graphical model
- Time-series
- Conditional independence

Example Machine Learning Tasks
- Classification: Discriminative modelling
- Classification: Generative modelling
- Density estimation
- Discrete Bayesian Networks
- Markov model

Algorithms for latent variable models
- Gradient algorithms
- Expectation maximisation
- Bayesian Inference

Coding Exercises
- Density estimation
- Classification

Summary
- Summary

Notational problems

In probability theory, we are dealing with functions P on sets, called measures. In statistics, we are dealing with random variables. While these two can be related, it is common to use shorthand notation in statistics. If in doubt, use the following conversion table.

Probability theory

1. Sets $A \subset \Omega$ are events.
2. Probability measure P on Ω.
3. $P(A) \in [0, 1]$ with $A \subset \Omega$
4. $P(A \cap C)$ the probability $\omega \in A \cap C$
5. $P(A|B)$ the probability of A if B is true.
6. Marginal distribution $P(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap H_i)$
7. Bayes's theorem: $P(A \mid B) = P(A \cap B)/P(B)$.

Statistics

1. Random variables $x : \Omega \to X$.
2. The measure P is implicit.
3. $\mathbb{P}(x)$ for the distribution of x
4. $\mathbb{P}(x, y)$ the joint distribution of x, y.
5. $\mathbb{P}(x|y)$ the distribution of x if y is given.
6. Marginal distribution $\mathbb{P}(y) = \sum_{i \in X} \mathbb{P}(x = i, y)$.
7. Bayes's theorem: $\mathbb{P}(x|y) = \mathbb{P}(x, y)/\mathbb{P}(y)$

Probability and statistics

Probability theory

Here, we talk a lot about probability measures P on some space Ω. These are functions on sets, so that

- $P(\Omega) = 1$
- $P(A) \in [0, 1]$ for any $A \subset \Omega$
- $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$.

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Notes on probability.

Measures: Functions on sets.

The idea of probability is that it is a type of measure. Measures in everyday life include length, mass, or area. The bigger the 'area' of a set, the more likely it is. It also has the same properties as areas, meaning that the total area of two separate sets is their joint area.

Subsets of Ω

When Ω is a finite set, we can consider any subset $A \subset \Omega$. However, in general we are restricted to an *algebra* of subsets. However, this is just a technical issue.

Statistics

In statistics, we are interested in random variables $x, y, \ldots$ which are functions $x : \Omega \to X$, $y : \Omega \to Y$ etc.

- We write $\mathbb{P}(x)$ as a shorthand for *the probability distribution of* x.
- The distribution of x arises in the following way:
  1. Sample ω from P
  2. Calculate $x(\omega)$.
- We can get a probability measure P_x by measuring how often x falls in various sets $A \subset X$:

$$P_x(A) \triangleq P(\{\omega : x(\omega) \in A\})$$

Here, $\{\omega : x(\omega) \in A\}$ is literally the set of random outcomes ω for which $x(\omega)$ is in the set A. To avoid writing this cumbersome expression, we simply write $\mathbb{P}(x)$ as a general shorthand. This allows us to also write

$$\mathbb{P}(x \in A) = P_x(A).$$

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Notes on random variables. The previous slide shows how we can get probability measures P_x for random variables x. It sometimes is necessary to think about the actual measure P_x, but we usually do not have to assign probabilities to sets! We can just say $\mathbb{P}(x)$ loosely, without specifying whether or not we are talking about the probability of a set, or even just the related density.

Joint distributions, multiple variables

It is still possible to define a probability measure for multiple variables. Typically, it is done as follows.

- Let $x : \omega \to X$, $y : \omega \to Y$ be two RVs.
- The measure $P_{x,y}$ has to be defined on the subsets of $X \times Y$.

$$P_{x,y}(A) = P(\{\omega : (x(\omega), y(\omega)) \in A\}$$

However, we usually just speak of the joint distribution $\mathbb{P}(x, y)$. If necessary, we may say $\mathbb{P}((x, y) \in A)$ for specific probabilities.

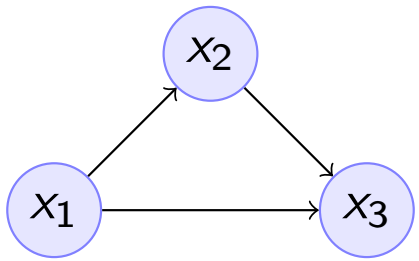
Factoring joint distributions.

For any sequence of random variables $x_1, \ldots, x_T$, their joint distribution can be written in terms of the factors

$$\begin{aligned}\mathbb{P}(x_1, \ldots, x_T) &= \mathbb{P}(x_1)\,\mathbb{P}(x_2|x_1)\,\mathbb{P}(x_3|x_2, x_1)\cdots\mathbb{P}(x_T|x_1, \ldots, x_{T-1}) \\ &= \mathbb{P}(x_1)\prod_{t=2}^{T}\mathbb{P}(x_t|x_1, \ldots, x_{t-1})\end{aligned}$$

However, these factors can be simplified if there only few dependencies between the random variables. The structure of dependencies can be formalised in terms of graphical models.

Graphical models



Directed acyclic graph

A directed acyclic graph with

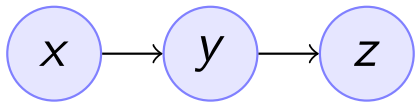
- Nodes $x_1, \ldots, x_n$
- Edges $x_i \to x_j$.

so that there are no cycles in the graph.

Conditional independence from graphical models

- Each node of the model corresponds to a random variable
- The parent of a node are the direct dependencies of the random variable.

Model specification



- The graphical model tells us what depends on what.
- It does not tell us how to generate data.
- We need to specify the functional relationship between variables.

Statistics

$$x \sim f$$
$$y \mid x = a \sim g(a)$$
$$z \mid y = b \sim h(b)$$

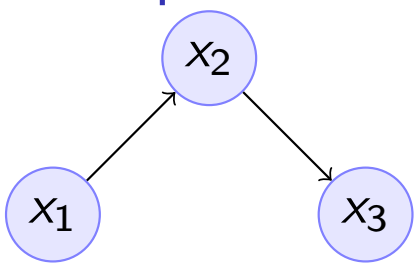
Python

```
x = f()
y = g(x)
z = h(y)
```

Graphical models and factorisation

- Graphical models tell us what directly depends on what
- They allow us to simplify the joint distribution of the variables
- This is formed into a product of factors, one for each variable.

Example



This graphical model implies the factorisation

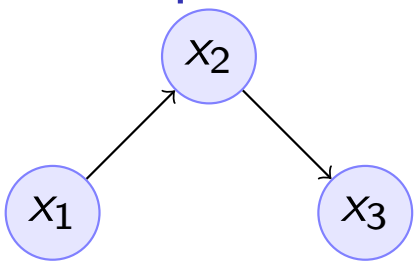
$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2, x_1)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1) = \mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$$

Notice that for each factor $\mathbb{P}(x_i \mid x_j)$, x_j is the parent of x_i.

Conditional independence (Chain)

- Graphical models tell us what directly depends on what
- Consequently, they also specify conditional independence

Example



$$\mathbb{P}(x_1, x_2, x_3) = \mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$$

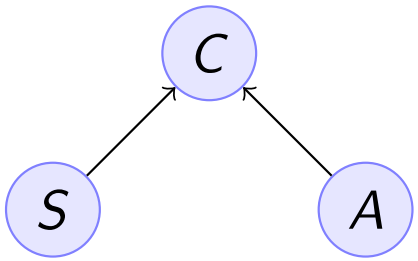
Definition (Conditional independence)

We say that variables x, y are conditionally independent given z and write $x \perp\!\!\!\perp y \mid z$ if and only if

$$\mathbb{P}(x, y \mid z) = \mathbb{P}(x \mid z)\,\mathbb{P}(y \mid z)$$

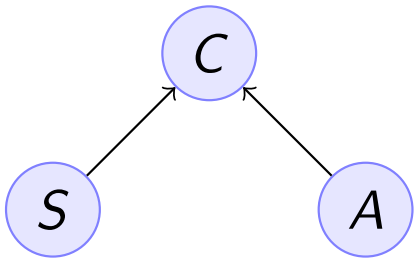
- In the above example, it holds that $x_3 \perp\!\!\!\perp x_1 \mid x_2$.

Smoking and lung cancer (Common effect)



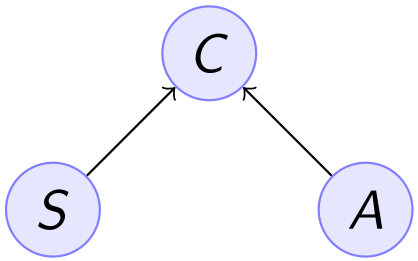
- Smoking and lung cancer graphical model.
- S: Smoking, C: cancer, A: asbestos exposure.

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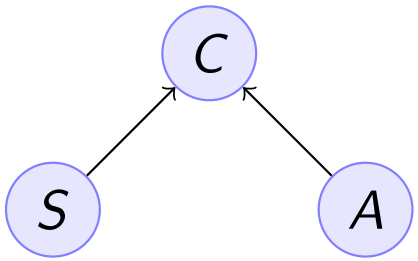


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XOR example

- C = xor(S, A).

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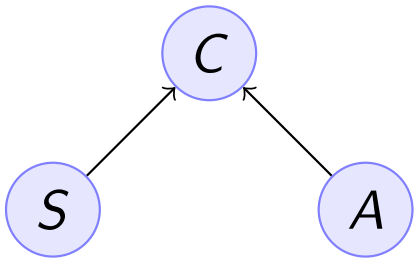


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- C = xor(S, A).
- If we know C = 1, S = 1, what is A?

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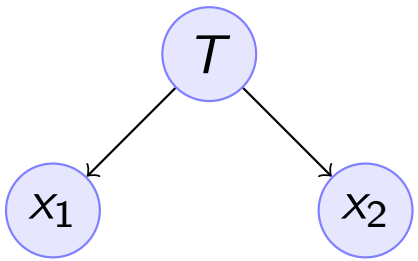


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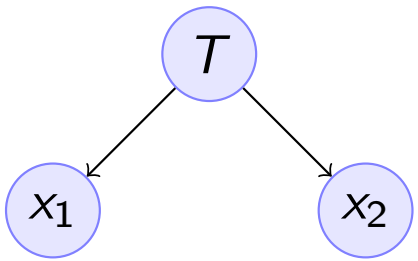
Time of arrival at work (Common Cause)



Time of arrival at work graphical model where T is a traffic jam and x_1 is the time John arrives at the office and x_2 is the time Jane arrives at the office.

- Even though x_1, x_2 are not independent, they become independent once you know T, i.e. $x_1 \perp\!\!\!\perp x_2 \mid T$.

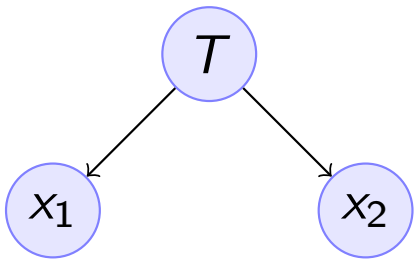
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- Proof:

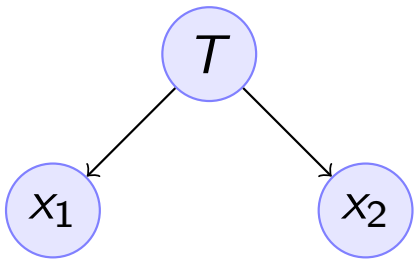
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- Proof:
- $\mathbb{P}(x_1, x_2, T) = \mathbb{P}(x_2 \mid T)\,\mathbb{P}(x_1 \mid T)\,\mathbb{P}(T)$ from the graph.

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- $\mathbb{P}(x_1, x_2, T)/\,\mathbb{P}(T) = \mathbb{P}(x_1, x_2 \mid T) = \mathbb{P}(x_2 \mid T)\,\mathbb{P}(x_1 \mid T)$. □

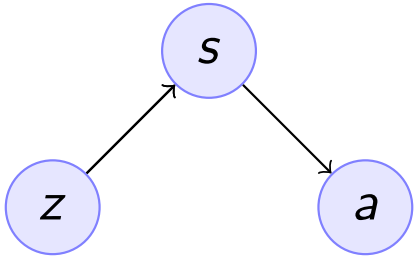
School admission

Example

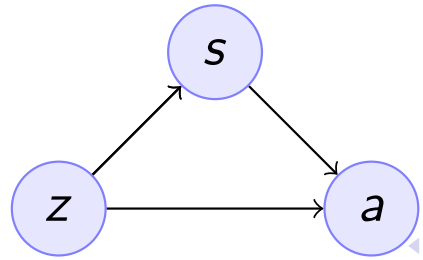
School	Male	Female
A	62	82
B	63	68
C	37	34
D	33	35
E	28	24
F	6	7
Average	50	27

- z: gender
- s: school applied to
- a: admission

- Admission rate (%) for each school.
- The average admission rate depends on the application rate!



Is admission



How about here?

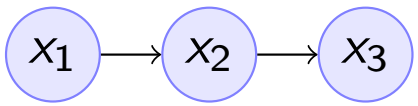
Time-series

A sequence of data $x_1, \ldots, x_t$, is called a time series. Examples include

- Text
- Speech
- Music
- Prices
- Weather

Generally, in a time series the future observations only depend on past ones. The general idea is captured by Markov models. Even though they appear simple, they are general enough to encompass all possible models for time series.

Markov chain



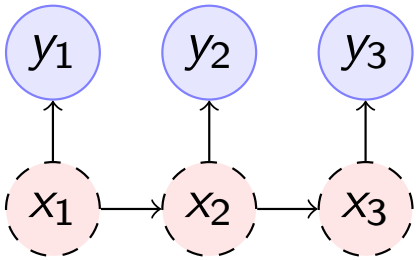
The Markov property

In a Markov chain, the observations form a chain. In general, the following independence condition holds:

$$\mathbb{P}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(x_{t+1} \mid x_t)$$

meaning that knowing x_t is sufficient to predict x_{t+1}, and the previous observations are not required.

Hidden Markov model



Hidden states

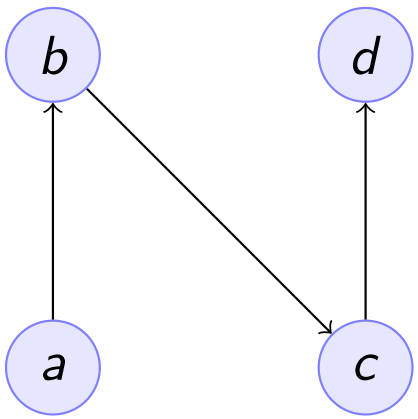
In a hidden Markov model, the observations y_t are not a chain. The variable x_t is not seen, and called a hidden state. The Markov property still holds for x_t,

$$\mathbb{P}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(x_{t+1} \mid x_t)$$

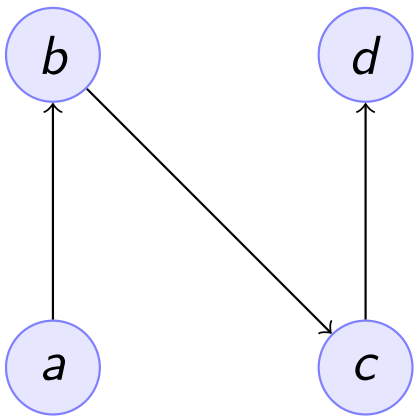
but not for y_t. However, if we know the current state, it is sufficient for us to predict the observations:

$$\mathbb{P}(y_t \mid y_{t-1}, \ldots, y_1, x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(y_t \mid x_t).$$

What is the model for this graph?

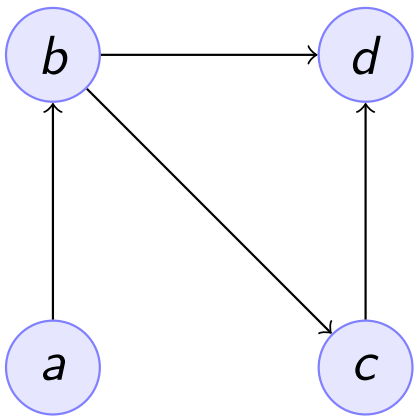


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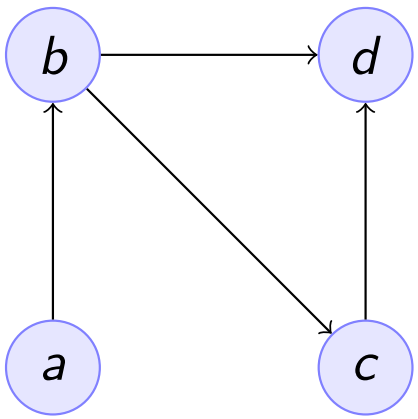


$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|c)\,\mathbb{P}(c|b)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

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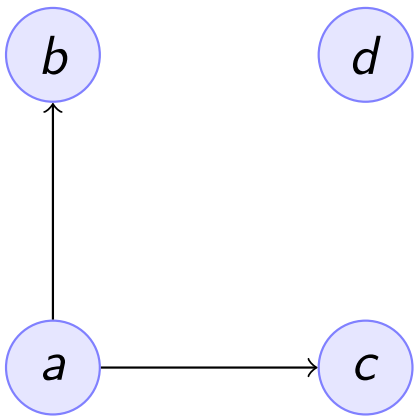


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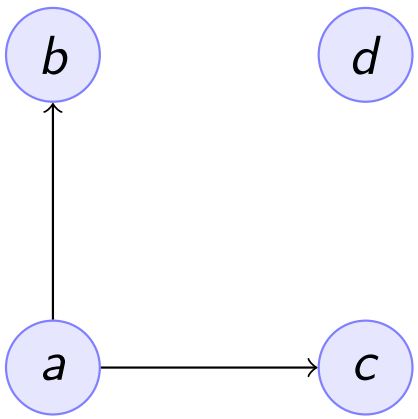


$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d|b, c)\,\mathbb{P}(c|b)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

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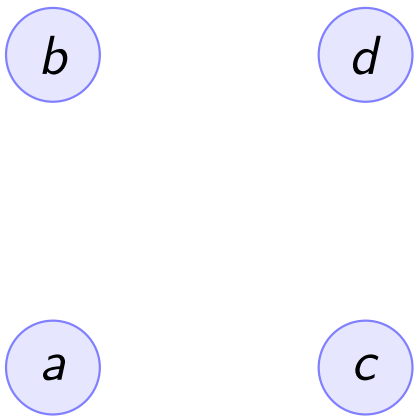


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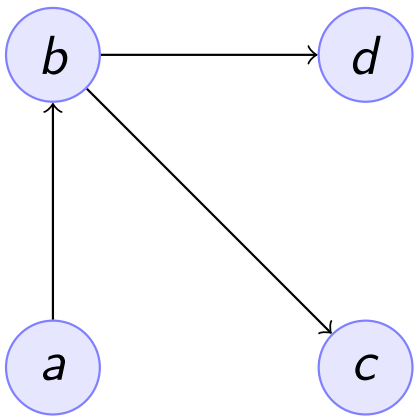
$$\mathbb{P}(a, b, c, d) = \mathbb{P}(d)\,\mathbb{P}(c|a)\,\mathbb{P}(b|a)\,\mathbb{P}(a)$$

Draw the graph for this model



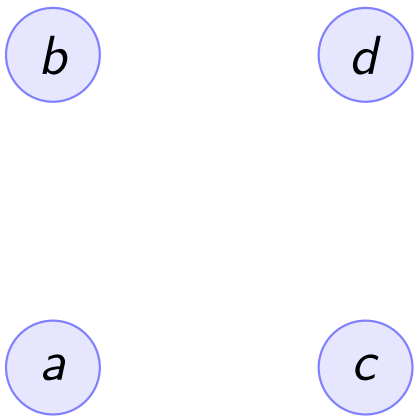
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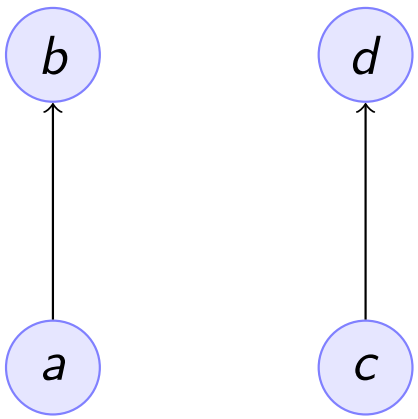
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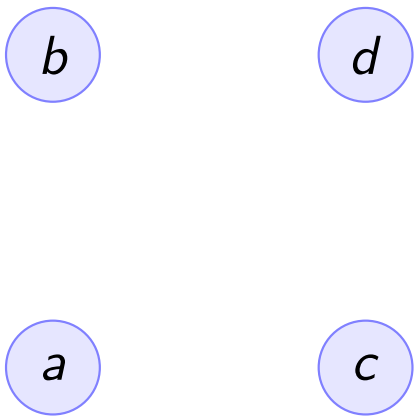
$$\mathbb{P}(a, b, c, d) = \mathbb{P}(a)\,\mathbb{P}(b|a)P(d|c)\,\mathbb{P}(c)$$

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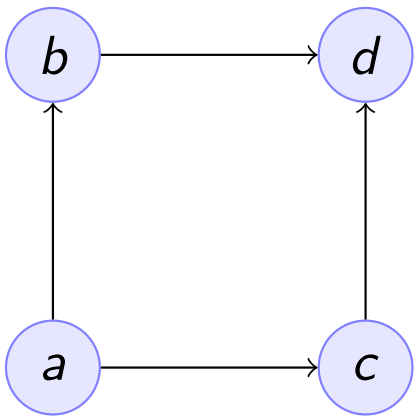
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Draw the graph for this model



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Conditional independence (general)

- Consider variables $x_1, \ldots, x_n$.
- Let B, D be subsets of $[n]$, and
- $\boldsymbol{x}_B \triangleq (x_i)_{i \in B}$ be the variables with indices in B.
- $\boldsymbol{x}_{-j} \triangleq (x_i)_{i \neq j}$ all the variables apart from x_j.

Definition (Conditional independence)

We say x_i is conditionally independent of $\boldsymbol{x}_B$ given $\boldsymbol{x}_D$ and write

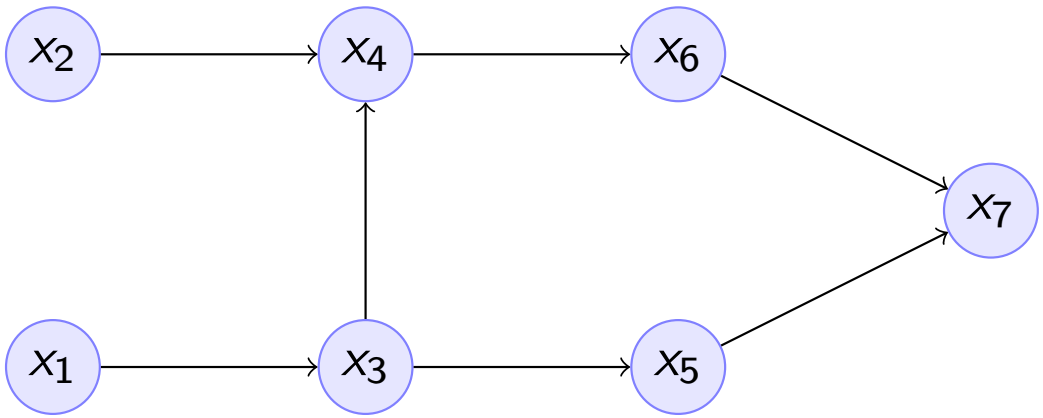
$$x_i \perp\!\!\!\perp \boldsymbol{x}_B \mid \boldsymbol{x}_D$$

if and only if:

$$\mathbb{P}(x_i, \boldsymbol{x}_B \mid \boldsymbol{x}_D) = \mathbb{P}(x_i \mid \boldsymbol{x}_D)\, \mathbb{P}(\boldsymbol{x}_B \mid \boldsymbol{x}_D).$$

- For this to hold in graphical model, D must separate i from B in the graph.

More complex example



In this example, we have:

$$x_7 \perp\!\!\!\perp x_1, x_2 \mid x_3, x_4$$

and

$$x_7 \perp\!\!\!\perp x_3 \mid x_4, x_5$$

Graphical models

Example Machine Learning Tasks
- Classification: Discriminative modelling
- Classification: Generative modelling
- Density estimation
- Discrete Bayesian Networks
- Markov model

Algorithms for latent variable models

Coding Exercises

Summary

Discriminative modelling

General idea

- Data (x_t, y_t).
- Need to model $\mathbb{P}(y|x)$.
- No need to model $\mathbb{P}(x|y), \mathbb{P}(y)$.

Examples

- Logistic regression.
- k-Nearest Neighbour
- Multi-Layer Perceptron

Discriminative modelling

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Examples
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- k-Nearest Neighbour
- Multi-Layer Perceptron

What we model. Of course, if we want to model the distribution $\mathbb{P}(x, y)$ and x, y are not independent, then it should not matter whether or not we factorise $\mathbb{P}(x|y)\,\mathbb{P}(y)$ or $\mathbb{P}(x)\,\mathbb{P}(y|x)$.

However, in order for us to perform classification or regression, we only need $\mathbb{P}(y|x)$.

Then there is no need to model $\mathbb{P}(x)$ explicitly.

Generative modelling

General idea

- Data (x_t, y_t).
- Need to model $\mathbb{P}(y|x)$.
- Model the complete data distribution: $\mathbb{P}(x|y)$, $\mathbb{P}(x)$, $\mathbb{P}(y)$.
- Calculate $\mathbb{P}(y|x) = \frac{\mathbb{P}(x|y)\,\mathbb{P}(y)}{\mathbb{P}(x)}$.

Examples

- Naive Bayes classifier.
- Gaussian mixture model.

Modelling the data distribution in classification

- Need to estimate the density $\mathbb{P}(x|y)$ for each class y.
- Need to estimate $\mathbb{P}(y)$.

Generative modelling

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Modelling the data distribution in classification
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- Need to estimate $\mathbb{P}(y)$.

Generative modelling. Generative modelling generally means modelling the joint distribution of all variables. So, if we are looking at the

The basic graphical model

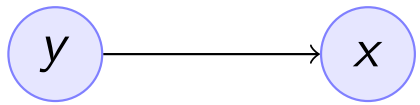
A discriminative classification model

Here $\mathbb{P}(y|x)$ is given directly.



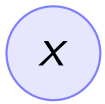
A generative classification model

Here $\mathbb{P}(y|x) = \mathbb{P}(x|y)\,\mathbb{P}(y)/\,\mathbb{P}(x)$.

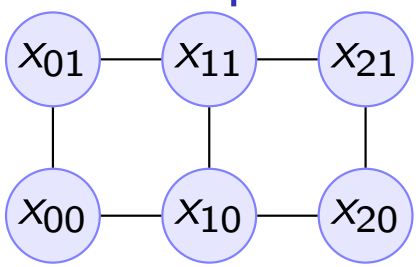


An unsupervised generative model with no structure

Here we just have $\mathbb{P}(x)$.



An unsupervised image model at the pixel level

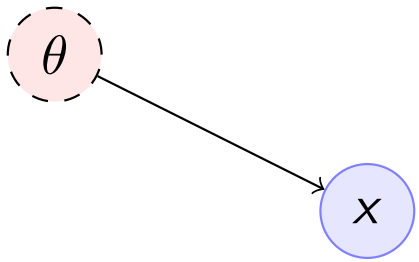


Here we model the factors $\mathbb{P}(x_{i,j}|x_{i-1,j}, x_{i,j-1}, x_{i+1,j}, x_{i,j+1})$

Adding parameters to the graphical model

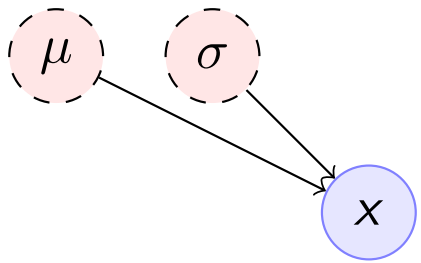
- We can also see the parameters of the distribution as (random) variables.
- We can put those random variables in the graphical model as well.
- Since the parameters are not observed, we denote them with dashed circles.
- They are a type of latent or hidden variable.

A Bernoulli RV



Here, $x|\theta \sim \text{Bernoulli}(\theta)$

A normally distributed variable



Here $x|\mu, \sigma \sim \text{Normal}(\mu, \sigma^2)$

Classification: Naive Bayes Classifier

- Data (x, y)
- $x \in X$
- $y \in Y \subset \mathbb{N}$, N_i: amount of data from class i.

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Separately model each class

- Assume each class data comes from a different normal distribution
- $x|y = i \sim \text{Normal}(\mu_i, \sigma_i I)$
- For each class, calculate
  - Empirical mean $\hat{\mu}_i = \sum_{t:y_t=i} x_t / N_i$
  - Empirical variance $\hat{\sigma}_i$.

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  - Empirical variance $\hat{\sigma}_i$.

Decision rule

Use Bayes's theorem:

$$\mathbb{P}(y|x) = \mathbb{P}(x|y)\,\mathbb{P}(y)/\,\mathbb{P}(x),$$

choosing the y with largest posterior $\mathbb{P}(y|x)$.

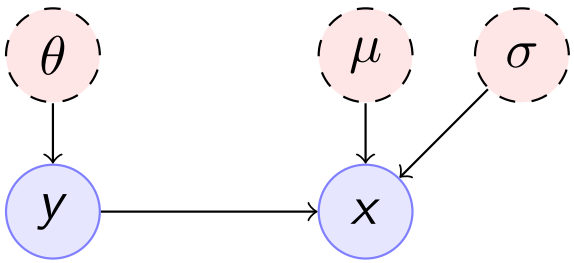
- $\mathbb{P}(x|y = i) \propto \exp(-\|\hat{\mu}_i - x\|^2/\hat{\sigma}_i^2)$

Graphical model for the Naive Bayes Classifier

When $x \in \mathbb{R}$

Assume k classes, then

- $\mu = (\mu_1, \ldots, \mu_k)$
- $\sigma = (\sigma_1, \ldots, \sigma_k)$
- $\theta = (\theta_1, \ldots, \theta_k)$



- $y \mid \theta \sim \text{Mult}(\theta)$
- $x \mid y, \mu, \sigma \sim \text{Normal}(\mu_y, \sigma_y^2)$

Density estimation

The simplest type of generative model is just modelling the distribution of x. There are a number of models for this.

Parametric models

- Fixed histograms
- Gaussian Mixtures

Non-parametric models

- Variable-bin histograms
- Infinite Gaussian Mixture Model
- Kernel methods

Histograms

Fixed histogram

- Hyper-Parameters: number of bins
- Parameters: Number of points in each bin.

Variable histogram

- Hyper-parameters: Rule for constructing bins
- Generally $\sqrt{n}$ points in each bin.

Gaussian Mixture Model

Hyperparameters:

- Number of Gaussian k.

Parameters:

- Multinomial distribution θ over Gaussians
- For each Gaussian i, center μ_i, covariance matrix Σ_i.

Algorithms:

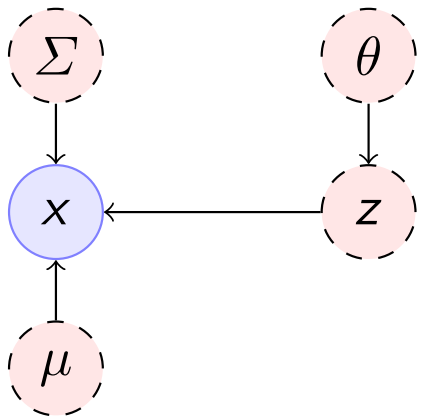
- Expectation Maximisation
- Gradient Ascent
- Variational Bayesian Inference (with appropriate prior)

Details of Gaussian mixture models

Model. For each point x_t:

- $z_t \mid \theta \sim \text{Mult}(\theta_i)$, $\theta \in \mathbb{\Delta}^k$
- $x_t | z_t = i \sim \text{Normal}(\mu_i, \Sigma_i)$.
- $\text{Mult}(\theta)$ is multinomial

$$\mathbb{P}(z_t = i \mid \theta) = \theta_i$$



- $\text{Normal}(\mu, \Sigma)$ is multivariate Gaussian

$$p(x \mid \mu, \Sigma) \propto \exp(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu))$$

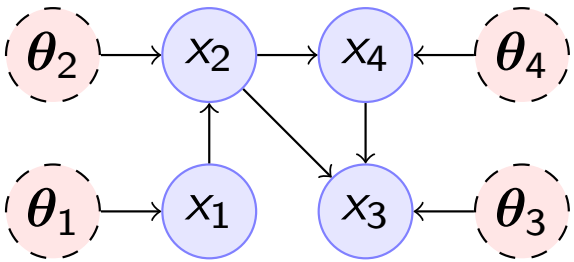
- The generating distribution is

$$p(x|\theta, \mu, \Sigma) = \sum_{z \in [k]} p(x \mid \mu_z, \Sigma_z)\, \mathbb{P}(z \mid \theta).$$

Applications of Gaussian mixture models

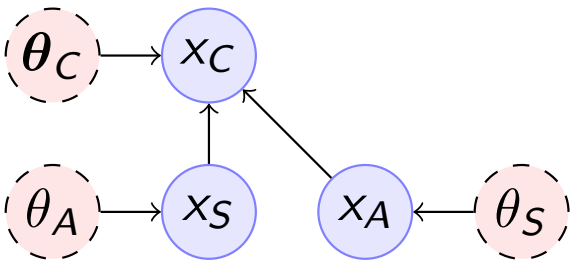
- Density estimation
- Clustering
- Used as part of a more complex model.

Discrete Bayesian Network



- A directed acyclic graph (DAG) defined on variables $x_1, \ldots, x_n$ with each x_n taking a finite number of values,
- Let S_i be the indices corresponding to parent variables of x_i.
- $x_i \mid \boldsymbol{\theta}_i, x_{S_i} = k \sim \mathrm{Mult}(\boldsymbol{\theta}_{i,k})$.

Example: Lung cancer, smoking and asbestos

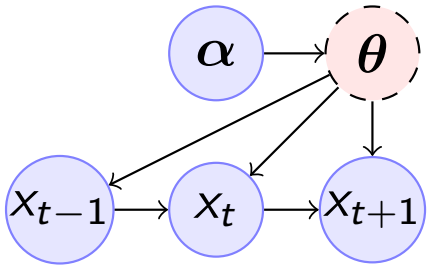


$$P_{\theta_A}(x_A = 1) = \theta_A \quad (1)$$

$$P_{\theta_S}(x_S = 1) = \theta_S \quad (2)$$

$$P_{\boldsymbol{\theta}_C}(x_C = 1 \mid X_A = j, X_S = k) = \theta_{C,j,k} \quad (3)$$

Finite Markov models



A Markov model obeys

$$\mathbb{P}_{\boldsymbol{\theta}}(x_{k+1}|x_k,\ldots,x_1) = \mathbb{P}_{\boldsymbol{\theta}}(x_{k+1}|x_k)$$

i.e. the graphical model is a chain. We are usually interested in homogeneous models, where

$$\mathbb{P}_{\boldsymbol{\theta}}(x_{k+1} = i \mid x_k = j) = \theta_{i,j} \qquad \forall k$$

Inference for finite Markov models

Finite Markov models are special cases of discrete Bayesian networks: we can consider each random variable x_{t+1} as having a Multinomial distribution, whose parameter depends on what the value of x_t. So, when $x_t = i$, then x_{t+1} is multinomial with parameter $\boldsymbol{\theta}_i$.

Inference for the Multinomial distribution

The Multinomial distribution

If $x_t \mid \boldsymbol{\theta} \sim \mathrm{Mult}(\boldsymbol{\theta})$ then $\mathbb{P}(x_t = i \mid \boldsymbol{\theta}) = \theta_i$.

The Dirichlet distribution

The conjugate prior for a Multinomial parameter is the Dirichlet distribution, which is the multivariate extension of the Beta. Its density is $p(\boldsymbol{\theta} \mid \boldsymbol{\alpha}) \propto \prod_{i=1}^{n} \theta_i^{\alpha_i - 1}$.

The Dirichlet-Multinomial conjugate pair

- If $x_t \in [n]$ then $x_{t+1} \mid \boldsymbol{\theta}, x_t = i \sim \mathrm{Mult}(\boldsymbol{\theta}_i)$, $\boldsymbol{\theta}_i \in \mathbb{\Delta}^n$
- Prior $\boldsymbol{\theta}_i \mid \boldsymbol{\alpha} \sim \mathrm{Dir}(\boldsymbol{\alpha})$ for all $i \in [n]$.
- Posterior $\boldsymbol{\theta}_i \mid x_1, \ldots, x_t, \boldsymbol{\alpha} \sim \mathrm{Dir}(\boldsymbol{\alpha}^{(t)})$ with

$$\alpha_{i,j}^t = \alpha_{i,j} + \sum_{k=1}^{t} \mathbb{I}\left\{x_k = i \wedge x_{k+1} = j\right\}, \qquad \boldsymbol{\alpha}^0 = \boldsymbol{\alpha}.$$

As you can see, the parameters of the Dirichlet distribution are updated the same way as those of the beta: simply increase the counts of the observed outcomes.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models
- Gradient algorithms
- Expectation maximisation
- Bayesian Inference

Coding Exercises

Summary

Gradient ascent

In the following we use θ for all the parameters of the Gaussian mixture model, with $x = (x_1, \ldots, x_T)$ and $z = (z_1, \ldots, z_T)$

Objective function

One way to estimate θ is through maximising the likelihood $L(\theta) = \mathbb{P}(x|\theta)$

Marginalisation over latent variable

However, we need to marginalise over all values z

$$L(\theta) = \sum_z \mathbb{P}(z, x|\theta)$$

For T data points and k different values of z_t, there are k^T vectors z to sum over.

Gradient ascent

If we can calculate the gradient of L, we can use gradient ascent to update our parameters:

$$\theta^{(n+1)} = \theta^{(n)} + \alpha \nabla_\theta L(\theta).$$

Gradient calculation (method I)

Here we use the log trick: $\nabla \ln f(\theta) = \nabla f(\theta)/f(\theta)$.

$$\nabla_\theta L(\theta) = \sum_z \nabla_\theta \mathbb{P}(z, x \mid \theta) \tag{4}$$

$$= \sum_z \mathbb{P}(z, x \mid \theta) \nabla_\theta \ln \mathbb{P}(z, x \mid \theta) \tag{5}$$

$$= \sum_z \mathbb{P}(x \mid z, \theta)\, \mathbb{P}(z \mid \theta) \nabla_\theta \ln \mathbb{P}(z, x \mid \theta) \tag{6}$$

We can approximate the sum by sampling $z^{(i)}$ to obtain:[1]

$$\frac{1}{m} \sum_{i=1}^{m} \mathbb{P}(x \mid z^{(i)}, \theta) \nabla_\theta \ln \mathbb{P}(z^{(i)}, x \mid \theta), \qquad z^{(i)} \sim \mathbb{P}(z \mid \theta)$$

We can use either the exact gradient or the stochastic estimate to perform (stochastic) gradient ascent.

[1] Not really necessary for GMMs, but needed for other latent variable models.

Gradient calculation (method II)

The problem with the previous method is that we sample from the prior $\mathbb{P}(z \mid \theta)$. Let us now start with the log-likelihood and use the log trick twice

$$\nabla_\theta \ln \mathbb{P}(x \mid \theta) = \frac{\nabla \mathbb{P}(x \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{7}$$

$$= \frac{\sum_z \nabla \mathbb{P}(x, z \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{8}$$

$$= \frac{\sum_z \mathbb{P}(x, z \mid \theta) \nabla \ln \mathbb{P}(x, z \mid \theta)}{\mathbb{P}(x \mid \theta)} \tag{9}$$

$$= \sum_z \mathbb{P}(z \mid x, \theta) \nabla \ln \mathbb{P}(x, z \mid \theta). \tag{10}$$

- Methods I, II have the same stationary points.
- However, since method I ascends the likelihood, and method II the log-likelihood, performance is better with method II for large T, as method I's gradient is vanishingly small.
- Method II gives you lower-variance Monte-Carlo estimates if you sample (for more complex models), because you use $\mathbb{P}(z \mid x, \theta)$, which is more informative, though harder for complex models.

Gradient components of GMMs

Now that we looked at general latent variable models, let us see what we need to calculate for GMMs. Our parameters are $\theta = (\theta, \mu, \sigma)$.

$$\nabla_\theta \ln \mathbb{P}(z, x \mid \theta) = \sum_{t=1}^{T} \nabla_\theta \ln \mathbb{P}(z_t, x_t \mid \theta) = \sum_{t=1}^{T} \nabla_\theta \ln [\mathbb{P}(z_t \mid \theta)\, \mathbb{P}(x_t \mid z_t, \mu, \sigma)]$$

The individual terms are:

$$\nabla_\theta \ln \mathbb{P}(z_t|\theta) = \frac{1}{\mathbb{P}(z_t|\theta)} \nabla_\theta \mathbb{P}(z_t|\theta), \qquad \frac{\partial}{\partial \theta_i} = \frac{\mathbb{I}\{z_t = i\}}{\mathbb{P}(z_t|\theta)}$$

$$\frac{\partial}{\partial \mu_i} \ln \mathbb{P}(x_t \mid z_t = i, \mu, \sigma) = -\frac{\partial}{\partial \mu_i} \frac{(x_t - \mu_i)^2}{2\sigma_i^2} = \sigma_i^{-2}(x_t - \mu_i)$$

So the algorithm

- Randomly assigns z_t according to the current θ estimate
- Moves the centers towards the mean indicated by the z.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models
- Gradient algorithms
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Coding Exercises

Summary

A lower bound on the likelihood

For any distribution $G(z)$, and specifically for $G(z) = \mathbb{P}(z|x, \theta^{(k)})$:

$$\begin{aligned}
\ln \mathbb{P}(x|\theta) &= \sum_z G(z) \ln \mathbb{P}(x|\theta) = \sum_z G(z) \ln[\mathbb{P}(x, z|\theta)/\, \mathbb{P}(z|x, \theta)] \\
&= \sum_z G(z)[\ln \mathbb{P}(x, z|\theta) - \ln \mathbb{P}(z|x, \theta)] \\
&= \sum_z G(z) \ln \mathbb{P}(x, z|\theta) - \sum_z G(z) \ln \mathbb{P}(z|x, \theta) \\
&= \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta) \\
&\geq \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta^{(k)}) \\
&= Q(\theta \mid \theta^{(k)}) + \mathbb{H}(z \mid x, \theta^{(k)}),
\end{aligned}$$

where

$$\mathbb{H}(z \mid x, \theta^{(k)}) = -\sum_z \mathbb{P}(z \mid x, \theta^{(k)}) \ln \mathbb{P}(z \mid x, \theta^{(k)})$$

is the entropy of z for a fixed $x, \theta^{(k)}$. As this is not negative, $\ln \mathbb{P}(x|\theta) \geq Q(\theta \mid \theta^{(k)})$.

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For any distribution $G(z)$, and specifically for $G(z) = \mathbb{P}(z|x, \theta^{(k)})$:

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&= \sum_z G(z)[\ln \mathbb{P}(x, z|\theta) - \ln \mathbb{P}(z|x, \theta)] \\
&= \sum_z G(z) \ln \mathbb{P}(x, z|\theta) - \sum_z G(z) \ln \mathbb{P}(z|x, \theta) \\
&= \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta) \\
&\geq \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(x, z|\theta) - \sum_z \mathbb{P}(z|x, \theta^{(k)}) \ln \mathbb{P}(z|x, \theta^{(k)}) \\
&= Q(\theta \mid \theta^{(k)}) + \mathbb{H}(z \mid x, \theta^{(k)}),
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where

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is the entropy of z for a fixed $x, \theta^{(k)}$. As this is not negative, $\ln \mathbb{P}(x|\theta) \geq Q(\theta \mid \theta^{(k)})$.

Some information theory. Information theory notation can be a bit confusing. Sometimes we talk about random variables ω, and sometimes about probability measures P. This is context-dependent.

Entropy

For a random variable ω under distribution P, we denote the entropy as

$$\mathbb{H}_P(\omega) \equiv \mathbb{H}(P) \equiv \mathbb{H}(\omega) = -\sum_{\omega \in \Omega} P(\omega) \ln P(\omega).$$

KL Divergence

For two probabilities P, Q over random outcomes in the same space Ω, we define

$$D_{KL}(P \| Q) = \sum_{\omega \in \Omega} P(\omega) \ln \frac{P(\omega)}{Q(\omega)}$$

The Gibbs Inequality

$D_{KL}(P \| Q) \geq 0$, or $\sum_x \ln P(x) P(x) \geq \sum_x \ln Q(x) P(x)$.

From Gradient Method II to Eexpectation Maximisation

we showed that

$$\ln \mathbb{P}(x \mid \theta) \geq \sum_z \mathbb{P}(z \mid x, \theta^{(k)}) \ln \mathbb{P}(x, z \mid \theta).$$

for some fixed $\theta^{(k)}$.

- The idea of the Expectation Maximisation algorithm is, instead of maximising the log-likelihood, to maximise a lower bound on it.
- In this case, this is achieved by fixing the latent distribution.
- This makes the optimisation problem simpler, and we can find an exact solution (at least for some models).

EM Algorithm (Dempster et al, 1977)

- Initial parameter $\boldsymbol{\theta}^{(0)}$, observed data x
- For $k = 0, 1, \ldots$

– Expectation step:

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(k)}) \triangleq \mathbb{E}_{z \sim \mathbb{P}(z|x,\boldsymbol{\theta}^{(k)})}[\ln \mathbb{P}(x, z|\boldsymbol{\theta})] = \sum_z [\ln \mathbb{P}(x, z|\boldsymbol{\theta})] \, \mathbb{P}(z \mid x, \boldsymbol{\theta}^{(k)})$$

– Maximisation step:

$$\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}).$$

See *Expectation-Maximization as lower bound maximization, Minka, 1998*

Minorise-Maximise

EM can be seen as a version of the minorise-maximise algorithm

- $f(\boldsymbol{\theta})$: Target function to maximise
- $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$: surrogate function

Q Minorizes f

This means surrogate is always a lower bound so that

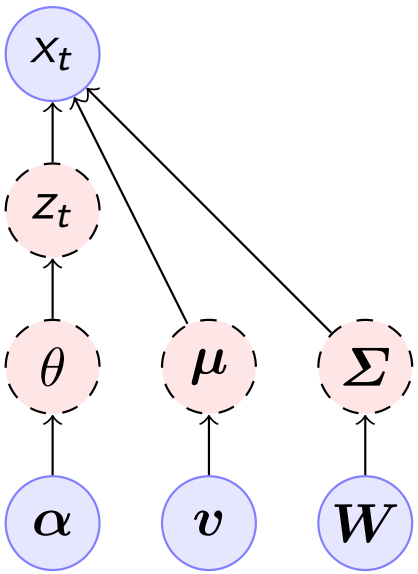
$$f(\boldsymbol{\theta}) \geq Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}), \qquad f(\boldsymbol{\theta}^{(k)}) = Q(\boldsymbol{\theta}^{(k)}|\boldsymbol{\theta}^{(k)}),$$

Algorithm

- Calculate: $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$
- Optimise: $\boldsymbol{\theta}^{(k+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$.

Bayesian inference for the GMM

We saw, for the Beta-Bernoulli and Dirichlet-Multinomial models, that a conjugate prior turns Bayesian inference into a simple parameter update. The same idea extends to the GMM, by placing a prior on every parameter.



$$\mu_i \mid \boldsymbol{v} \sim \text{Normal}(\boldsymbol{v}, I), \tag{11}$$
$$\Sigma_i \mid \boldsymbol{W} \sim \text{Wish}(\boldsymbol{W}), \tag{12}$$
$$\theta \mid \boldsymbol{\alpha} \sim \text{Dir}(\boldsymbol{\alpha}), \tag{13}$$
$$z_t \mid \theta \sim \text{Mult}(\theta), \tag{14}$$
$$x_t \mid \mu, \Sigma, z_t = i \sim \text{Normal}(\mu_i, \Sigma_i). \tag{15}$$

Exercise

Which of the five lines above, if any, would have a closed-form conjugate update if all the others were held fixed? Compare with the Beta-Bernoulli and Dirichlet-Multinomial cases covered earlier.

Graphical models

Example Machine Learning Tasks

Algorithms for latent variable models

Coding Exercises
- Density estimation
- Classification

Summary

GMM versus histogram

- Generate some data x from an arbitrary distribution in $\mathbb{R}$.
- Fit the data with a histogram for varying numbers of bins
- Fit a GMM with varying numbers of Gaussians
- What is the best fit? How can you measure it?

GMM Classifier

Base class: sklearn GaussianMixtureModel

- *fit()* only works for Density Estimaiton
- *predict()* only predicts cluster labels

Problem

- Create a GMMClassifier class
- *fit()* should take X, y, arguments
- *predict()* should predict class labels
- Hint: Use $predict_{proba}()$ and multiple GMM models

Summary

- CPTs store conditional probabilities for discrete random variables.
- Conditional independence reduces the number of parameters.
- Graphical models represent specific conditional independence structures.
- Models can also contain *latent* variables.
- Latent variables can be parameters, cluster IDs, etc

Example graphical models

- Naive Bayes
- Gaussian Mixture Model
- Linear regression
- Logistic regression
- Many other neural networks (those with a probabilistic interpretation).

Next: fairness!